Log Book Entry 3.1

NB2420 Electronic Instrumentation · Manual 3.1 · Aliasing and anti-aliasing

Hand in via BrightSpace

Name(s): Partner:

Date: Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

PREDICT AND MEASURE THE ALIAS

Tasks A1, A2, I1, I2

Sampling at $f_s = 1000$ Hz, so $f_N = 500$ Hz. Predict each FFT peak as homework, then read the measured peak off your own spectrum.

True frequency	Predicted FFT peak A1	Phase sign	Measured FFT peak I1
100 Hz			
450 Hz			
500 Hz			
650 Hz			
1100 Hz			

Square wave harmonics

Tasks A2 and I2: 100 Hz square, odd harmonics

Harmonic	Below, at or above f_N	Predicted position after sampling	Measured
100 Hz			
300 Hz			
500 Hz			
700 Hz			
900 Hz			

Anti-aliasing filter

Tasks A3 and I3: $R = 10$ kΩ, $f_c = 1/(2\pi RC)$

Capacitor	f_c calculated	Effect measured on the 700 and 900 Hz aliases	Suitable? Why
47 nF			
100 nF			

EQUIVALENT TIME SAMPLING AND CONCLUSIONS

Tasks I4, C1 to C3

Equivalent time sampling

Task I4: 73 Hz sine

Sampling rate used:

Points per reconstructed period:

Reconstructed frequency:

C1 Aliasing

What is aliasing and when does it occur? What is the highest component measurable at f_s? For a 500 Hz signal, which of 800, 1000, 1200 or 5500 Hz would you sample at, and why?

C2 Anti-aliasing

Explain how an RC low-pass filter prevents aliasing, and which capacitor you would pick for the Task I2 square wave.

C3 Equivalent time sampling

What must be true of the signal, what must be known about the timing, and what kind of event can never be reconstructed this way?

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off:

Date: